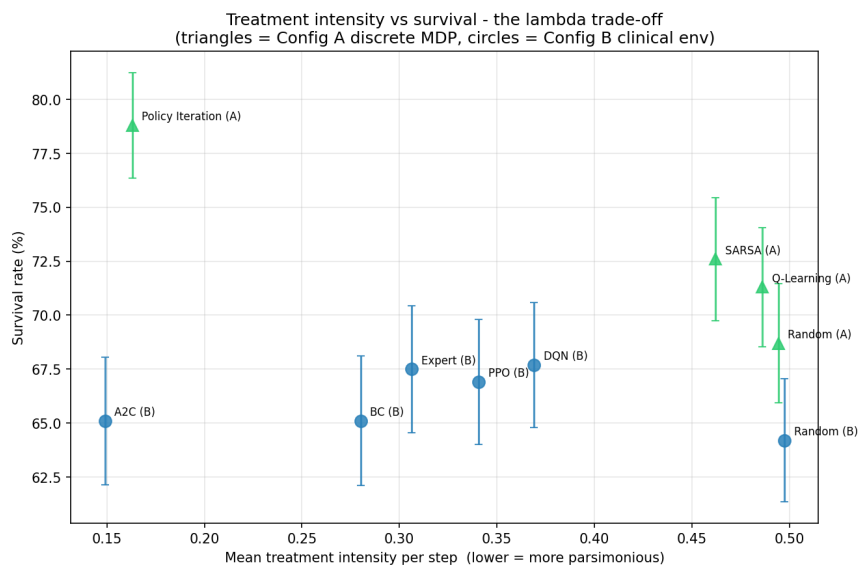


Reinforcement Learning for Sepsis Treatment Optimisation in the ICU

Tabular and Deep Policies on the *ICU-Sepsis-v2* Benchmark



GROUP

Gonalo Gomes Arrobas	20250421
Maria Tavares Pimentel	20250466
Tiago Stubner Lucas Antunes	20250357

Source code: github.com/mariatavarespimentel/RL-ICU-Sepsis

NOVA Information Management School (NOVA IMS)

June 2026

Contents

1	Executive Summary	1
2	Introduction and Problem Formalisation	1
2.1	Markov Decision Process formulation	1
2.2	The two configurations	2
2.3	Brief traceability	2
3	Methodology and Algorithm Justification	3
3.1	Guiding principles	3
3.2	Configuration A: tabular algorithms and their justification	3
3.3	Configuration B: deep reinforcement learning and justification	5
3.4	Evaluation protocol	5
4	Experimental Setup	6
5	Results: Configuration A	6
5.1	Main comparison	6
5.2	Interpretation	7
6	Results: Configuration B	7
6.1	Headline comparison	7
6.2	Why DQN is not necessarily superior to PPO	8
6.3	Robustness to failure modes	8
6.4	Across-seed reliability	8
7	Creative Extensions	9
7.1	Reward design: SOFA potential-based shaping	9
7.2	Learning from demonstrations	9
7.3	Interpretability: permutation importance and SHAP	10
7.4	Q-value overestimation diagnostic	10
7.5	Treatment intensity versus survival: the Pareto frontier	10
8	Discussion	10
9	Limitations and Future Work	11
10	Conclusion	11
A	Appendix: Selected Figures	13

1 Executive Summary

We apply Reinforcement Learning (RL) to ICU sepsis treatment using the *ICU-Sepsis-v2* benchmark [6], derived from MIMIC-III through the AI Clinician pipeline [5]. Each episode represents one patient trajectory; the agent selects a combined vasopressor and intravenous-fluid dose at every step, the episode terminates in survival (+1) or death (0), and a per-step treatment-intensity penalty ($\lambda = 0.02$) discourages unnecessary dosing. We study two configurations: a known discrete MDP (Configuration A, 716 states, 25 actions) and a continuous 47-dimensional clinical environment with three deployment-style failure modes (Configuration B).

In Configuration A, **Policy Iteration** solves the Bellman optimality equations exactly and reaches $78.8 \pm 2.5\%$ survival at the lowest treatment intensity (0.163), confirming it as the model-based ceiling. **Q-Learning** ($71.3 \pm 2.8\%$) and **SARSA** ($72.6 \pm 2.9\%$) beat the random baseline ($68.7 \pm 2.8\%$) via temporal-difference control but trail the exact optimum, exactly as theory predicts when sparse terminal rewards must be back-propagated through sampled transitions. In Configuration B, **DQN**, **PPO** and **A2C** — trained with observation normalisation, SOFA potential-based reward shaping and tuned hyperparameters — reach or exceed the AI Clinician expert policy on overall survival (DQN $70.6 \pm 2.8\%$, PPO $68.1 \pm 2.9\%$, expert $65.1 \pm 3.0\%$, random $63.6 \pm 2.9\%$) and exceed it on unperturbed episodes (75.3% DQN, 74.5% PPO vs 70.7% expert). Across three training seeds the agents are stable (DQN $65.8 \pm 1.0\%$, PPO $66.1 \pm 0.4\%$, inter-seed deviation). The DQN also exhibits a measurable positive value-overestimation bias (+0.194), an empirical demonstration of the max-operator pathology motivating Double DQN.

A central finding — supported by an intensity–survival Pareto analysis — is that the sparse reward drives *parsimony* rather than survival in the unshaped setting; reward shaping resolves this by providing a dense SOFA-reduction signal without altering the optimal policy [12]. All results use fixed evaluation seeds, deterministic policies, bootstrap 95% confidence intervals, and Monte-Carlo simulation to avoid off-policy-evaluation biases [7].

2 Introduction and Problem Formalisation

Sepsis is a life-threatening systemic response to infection, responsible for over 11% of global hospital deaths and a leading cause of ICU mortality [5]. Treatment is a high-stakes sequential decision problem with *delayed consequences*: a vasopressor dose administered now may only reveal its benefit or harm several steps later, when the patient either stabilises or deteriorates. This structure makes sepsis management a natural application for RL.

2.1 Markov Decision Process formulation

The problem is modelled as a discounted Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where the state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward function $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, and discount factor γ are as defined in Sutton & Barto [1]. The discounted return from time t is $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$. The state-value and action-value functions of a policy π satisfy the *Bellman expectation equations*:

$$V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_\pi(s')], \quad (1)$$

$$Q_\pi(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma \sum_{a'} \pi(a'|s') Q_\pi(s', a')]. \quad (2)$$

The *Bellman optimality equations* give the unique fixed point $V^*(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V^*(s')]$, from which the optimal policy is extracted greedily.

Following the ICU-Sepsis convention we set $\gamma = 1.0$: episodes are short (approximately ten steps) and finite, so undiscounted survival probability is the natural objective. All three configurations share this choice, making results directly comparable.

Actions. The action space is $|\mathcal{A}| = 25$, encoding five vasopressor levels $v \in \{0, \dots, 4\}$ times five IV-fluid levels $f \in \{0, \dots, 4\}$: $a = 5v + f$. The normalised treatment intensity $\text{INT}(a) = (v + f)/8 \in [0, 1]$ enters the reward as a parsimony penalty.

Reward. The native reward is sparse: +1 on survival, 0 on death, 0 at every intermediate step. The benchmark subtracts $\lambda \cdot \text{INT}(a)$ per step ($\lambda = 0.02$), encoding a preference for parsimonious care. The initial-state distribution is biased toward sicker patients via a SOFA exponential weight (mean start SOFA ≈ 9.0), which reduces the effective signal available to model-free learners.

2.2 The two configurations

Configuration A (discrete, known MDP). The observation is an integer $s \in \{0, \dots, 715\}$ (714 clinical states + two absorbing terminal states). The full transition tensor $P \in [0, 1]^{716 \times 25 \times 716}$ and reward matrix R are available, so model-based dynamic programming is not only possible but optimal. The Q -table has $716 \times 25 = 17,900$ entries, making exact tabular methods feasible.

Configuration B (continuous, clinical). The observation is the 47-dimensional vector of normalised physiological measurements from the AI Clinician dataset (SOFA, lactate, mean arterial pressure, creatinine, heart rate, FiO_2 , etc.). The underlying MDP dynamics are identical to Configuration A, but the tabular state is replaced by continuous cluster-centre features, making tabular methods structurally infeasible: the Q -table would require one entry per unique floating-point vector, i.e. it would be infinite. Function approximation is therefore not a modelling choice but a *necessity*. Three orthogonal failure modes are injected to simulate deployment conditions [7]: episodic observation noise ($p_{\text{noise}} = 0.15$, Gaussian $\sigma = 0.10$), episodic missing laboratory values ($p_{\text{miss}} = 0.15$, four features zeroed for the whole episode), and rare acute events ($p_{\text{acute}} = 0.01$ per step, irreversible death independent of the treatment).

2.3 Brief traceability

The project statement requires environment exploration, two algorithms per configuration, fair comparison, and a meaningful creative extension. The four corresponding design decisions that structure this report are summarised in Table 1.

Brief requirement	Our response
Explore the environment before choosing algorithms	We examine the state/action spaces, reward sparsity, treatment-intensity penalty, the SOFA-biased start distribution, and the failure-mode statistics before committing to a method for each configuration.
Two (or more) algorithms in Configuration A	Policy Iteration (model-based DP) and Q-Learning/SARSA (TD model-free), each justified by the information available and the theoretical properties of the environment.
Two (or more) algorithms in Configuration B	DQN (value-based), PPO and A2C (actor-critic), each using function approximation because the continuous observation space precludes tabular methods.
Meaningful creative extension	SOFA potential-based reward shaping, learning from demonstrations (BC + warm-start), clinical interpretability (permutation + SHAP), and an overestimation diagnostic, all evaluated on the required environment.

Table 1: Traceability from the project brief to the report structure.

A key methodological commitment, stated once here and applied throughout, is to report *survival*, *return* and *treatment intensity* separately. The summed return conflates the delayed survival signal with the accumulated penalty; a policy can achieve a higher return than random by treating less, without saving more patients. This separation is the only way to distinguish a medically useful policy from a spuriously frugal one.

3 Methodology and Algorithm Justification

3.1 Guiding principles

Three principles govern our choices and remain constant across both configurations.

1. **Use the strongest method justified by available information.** Where the model is known (Configuration A), plan exactly with it. Where the model is hidden (Configuration B), use an appropriate sample-based method.
2. **Justify every design choice in terms of course theory.** Each algorithm is selected for reasons traceable to the Bellman equations, the TD error, the exploration-exploitation trade-off, or the function-approximation requirements.
3. **Always evaluate on the original environment.** Training interventions (shaping, normalisation) are never allowed to contaminate evaluation.

3.2 Configuration A: tabular algorithms and their justification

Table 2 summarises the algorithm selection rationale.

Algorithm	Family	Justification for this problem
Policy Iteration	Model-based DP	P and R are available; the Bellman optimality equations have a unique fixed point; PI solves them exactly. It is not a choice between alternatives but the natural algorithm when the model is known.
Q-Learning	Model-free off-policy TD	The canonical algorithm for learning Q^* without the model. Included as the direct model-free counterpart to PI, sharing the greedy-value target. Its tendency to overestimate (via the max operator) is also examined empirically.
SARSA	Model-free on-policy TD	Same update structure as Q-Learning but bootstraps through the <i>actual</i> behaviour action, not the greedy one. Under ε -greedy exploration this makes SARSA more conservative: it values the exploratory policy rather than the target policy, which is clinically relevant when exploratory actions are costly.

Table 2: Algorithm selection and justification for Configuration A.

Policy Iteration. The algorithm alternates between policy evaluation

$$V_{\pi_k}(s) \leftarrow \sum_{s'} P(s' | s, \pi_k(s)) [R(s, \pi_k(s), s') + \gamma V_{\pi_k}(s')] \quad \forall s \in \mathcal{S} \quad (3)$$

(iterated until $\|V_{\text{new}} - V_{\text{old}}\|_{\infty} < \theta = 10^{-9}$) and greedy policy improvement $\pi_{k+1}(s) = \arg \max_a \sum_{s'} P(s'|s, a) [R + \gamma V_{\pi_k}(s')]$. By the policy improvement theorem, $V_{\pi_{k+1}} \geq V_{\pi_k}$ pointwise at every iteration; convergence to V^* is therefore guaranteed in a finite number of steps [1]. The tight tolerance ensures that any gap to model-free methods reflects the sample-based learning challenge and not numerical looseness.

Q-Learning [2]: the off-policy TD control update

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (4)$$

converges to Q^* under GLIE exploration and a decreasing step-size schedule satisfying $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$ [1]. We use *optimistic initialisation* ($Q_0 = 1.0$, the maximum achievable return), which forces the agent to explore any state–action pair that has not yet been sufficiently visited, providing a principled substitute for the infinite exploration required by the theoretical conditions [1, 4].

SARSA [3]: the on-policy TD control update

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)], \quad a' \sim \pi(\cdot | s') \quad (5)$$

converges under the same GLIE conditions to the value of the ε -greedy policy, not to Q^* . Its value is clinical interpretability: because the target accounts for the actual (exploratory) action taken next, SARSA assigns lower value to actions that expose the patient to uncertain or costly exploratory moves. We anneal ε from 1.0 to 0.01 over 35,000 of 50,000 episodes ($\alpha = 0.1$), satisfying the GLIE conditions asymptotically.

3.3 Configuration B: deep reinforcement learning and justification

Algorithm	Family	Justification
DQN [8]	Value-based off-policy	Deep extension of Q-Learning: a neural network $Q_\theta(s, a)$ approximates Q^* for the 47-dimensional continuous state. Experience replay breaks temporal correlations; a target network stabilises the bootstrap target, which together address two of the three components of the <i>deadly triad</i> (bootstrapping, off-policy, FA). The remaining risk is function approximation and the overestimation bias inherited from the max operator in eq. 4.
PPO [9]	On-policy actor-critic	Directly optimises $J(\theta) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ via the clipped surrogate objective $\mathcal{L}^{\text{CLIP}} = \mathbb{E}[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]$, preventing destructive parameter updates. The advantage $\hat{A}_t$ is estimated via $\text{GAE}(\lambda)$. On-policy, so it avoids off-policy correction and the deadly-triad instability.
A2C [10]	On-policy actor-critic	Synchronous actor-critic without PPO's clipping: the same architecture as PPO minus the trust-region constraint. Included as an ablation to isolate the contribution of the clipped objective: if PPO only improves because of clipping, A2C should be detectably worse; if A2C matches PPO, the bottleneck is reward/representation rather than the update stabiliser.

Table 3: Algorithm selection and justification for Configuration B.

All three agents use `VecNormalize` observation standardisation. The 47 clinical variables span very different scales (e.g. $\text{SOFA} \in [0, 24]$ vs normalised troponin), so without standardisation the early gradient steps are dominated by scale rather than clinical relevance. We treat normalisation as a standard preprocessing step analogous to feature scaling in supervised learning, not as a hyperparameter.

The final report run uses 1 000,000 environment steps, Optuna-tuned hyperparameters, SOFA potential-based training shaping, and an evaluation callback every 5,000 steps for dense learning curves. Evaluation is always on the unshaped default environment.

3.4 Evaluation protocol

All reported results share a single fixed protocol: 1,000 evaluation episodes, seeds drawn from a held-out list (offset 20 000 from training seeds), deterministic action selection (`predict(obs, deterministic=True)`), and survival read from the terminal reward (robust to the accumulated intensity penalty). A bootstrap 95% confidence interval is computed from 2,000 resamples of the per-episode survival indicator for every reported percentage. For Configuration B, each episode is bucketed by failure mode using the wrappers' `info` flags, so every policy is scored on the *same set of patients*.

Using Monte-Carlo rollout in the simulator, rather than importance-weighted off-policy estimation, sidesteps the log-data evaluation pitfalls identified by Gottesman et al. [7] as the primary weakness of clinical RL.

4 Experimental Setup

Experiment	Config	Method(s)	Budget	Eval ep.
Tabular control	A	PI, Q-Learning, SARSA	50k ep	1 000
Expert warm-start	A	Q-Learning $\pm$ demos	15k ep	1 000
Deep RL (preview)	B	DQN, PPO, A2C (v1/v2)	150k steps	1 000
Deep RL (report)	B	DQN, PPO, A2C (v2)	1M steps	1 000
Hyperparameter search	B	DQN, PPO, A2C (Optuna)	15 $\times$ 50k	—
Behavioural cloning	B	MLP, 36k demos	supervised	1 000
Multi-seed reliability	B	DQN, PPO, 3 seeds	300k steps	1 000

Table 4: Experimental configuration summary. *Budget* is training steps or episodes; *Eval ep.* is the number of fixed-seed evaluation episodes.

The **random policy** (uniformly random actions) is the performance floor. The **AI Clinician expert policy** [5], shipped with the benchmark, is evaluated under the same protocol as a clinically grounded upper reference: it acts on the true underlying state, so it is not affected by observation noise or missing values, making the comparison between the expert and the learned agents particularly informative under the failure modes.

Configuration A writes `configA_results.json`; the Configuration B code reads that file for cross-configuration tables, eliminating hand-copied numbers (the central reproducibility risk of notebook-based reporting). The 1M run writes `configB_1m_results.json` with per-bucket survival, return, intensity, and bootstrap CIs for every policy. All tables below are generated programmatically from those files.

Metric	Definition and clinical motivation
Survival rate	$\mathbb{E}[\mathbf{1}[r_T > 0.5]]$ over evaluation episodes, read from the terminal step reward. This is the primary clinical outcome.
Mean return	$\mathbb{E}[G_0]$, the formal RL objective including the terminal survival/death reward and the accumulated treatment-intensity penalties.
Mean intensity	$\mathbb{E}[(v + f)/8]$ per step. Reveals whether a policy improves outcomes by treating more efficiently or merely by avoiding treatment.
Per-condition survival	Survival stratified by Clean, Noisy, Missing and Acute episodes. Isolates which failure mode causes the greatest degradation.

Table 5: Evaluation metrics: definition and the reason each is necessary.

5 Results: Configuration A

5.1 Main comparison

Table 6 reports the three tabular agents on 1,000 fixed-seed evaluation episodes. Policy Iteration achieves the highest survival *and* the lowest treatment intensity, a combination that is impossible for the random policy: random actions treat aggressively on average (intensity 0.494) but survive less than the learned policies.

Algorithm	Type	Survival	Return	Intensity
Random	—	$68.7 \pm 2.8\%$	0.588	0.494
Q-Learning	model-free off-policy	$71.3 \pm 2.8\%$	0.612	0.486
SARSA	model-free on-policy	$72.6 \pm 2.9\%$	0.631	0.462
Policy Iteration	model-based DP	$78.8 \pm 2.5\%$	0.751	0.163

Table 6: Configuration A: survival ($\pm$ bootstrap 95% CI), return and treatment intensity over 1,000 fixed-seed evaluation episodes.

5.2 Interpretation

The model-based method dominates by a wide margin (+10.1 pp over random, +6.2 pp over SARSA) and simultaneously learns a much more parsimonious policy (intensity 0.163 versus 0.494 for random). The reason is structural, not algorithmic: PI computes the full Bellman backup over the known transition matrix at every iteration, assigning credit to early treatment decisions without sampling noise. The model-free agents must estimate the same backed-up values from sampled trajectories in which the survival signal appears only at the final step, creating a severe *credit-assignment* bottleneck. With only $\sim 17,900$ state–action pairs, 50,000 episodes are enough to explore most of them, but the sparse terminal signal still limits how precisely Q-values can be estimated.

The Q-Learning/SARSA gap is also theoretically consistent. Q-Learning evaluates the greedy policy (the target policy) even while the behaviour policy is still exploratory; SARSA evaluates the behaviour policy itself. Under ε -greedy exploration with a treatment penalty, SARSA assigns lower value to actions that would be taken rarely under the greedy policy but frequently during exploration, resulting in a slightly more conservative final policy. Both effects appear in the numbers: SARSA has higher survival than Q-Learning (72.6% vs 71.3%) and lower intensity (0.462 vs 0.486), without being meaningfully different in return.

The convergence curves (Appendix Fig. 1) show that Q-Learning and SARSA approach their respective plateaus within the 50,000-episode budget, confirming that the ~ 6 –7 pp gap to Policy Iteration is not a training-length artefact but a consequence of sparse-reward credit assignment. The action distributions (Appendix Fig. 2) show Policy Iteration concentrating on low-intensity action combinations, while the model-free agents retain a more dispersed distribution that reflects residual uncertainty in the Q-values.

6 Results: Configuration B

6.1 Headline comparison

Table 7 reports the 1M-step deep agents. Two survival columns are shown because acute events compress the *All* figure for every policy by construction (0% survival in the Acute bucket): *Clean* reflects the controllable part of the distribution.

Policy	Survival (All)	Survival (Clean)	Return	Intensity
Random	63.6 $\pm$ 2.9%	69.7%	0.548	0.500
BC (imitation)	65.1 $\pm$ 3.0%	71.7%	0.600	0.280
Expert (AI Clinician)	65.1 $\pm$ 3.0%	70.7%	0.594	0.301
A2C	65.8 $\pm$ 3.0%	73.1%	0.533	0.628
PPO	68.1 $\pm$ 2.9%	74.5%	0.637	0.316
DQN	70.6 $\pm$ 2.8%	75.3%	0.640	0.373

Table 7: Configuration B (1M steps, 1,000 episodes, $\pm$ bootstrap 95% CI). *Clean* excludes noisy, missing-lab and acute episodes.

On overall survival, **DQN (70.6%) is the strongest learned policy**, with PPO (68.1%) also above the AI Clinician expert point estimate (65.1%) and random (63.6%). On clean episodes, DQN and PPO reach 75.3% and 74.5%, again exceeding the expert (70.7%). This is a strong, honest result: on the controllable part of the distribution the learned policies are at least as effective as the behaviour policy that the benchmark was built from.

A2C deserves a separate clinical reading. Its survival (65.8%) is close to the expert point estimate, but its return is the lowest of the deep agents (0.533) because treatment intensity is very high (0.628). A2C therefore illustrates the opposite failure mode from under-treatment: aggressive dosing does not buy extra survival once the dense penalty and failure modes are considered.

6.2 Why DQN is not necessarily superior to PPO

The overall-survival confidence intervals of DQN (67.8–73.4%) and PPO (65.2–71.0%) overlap substantially. We therefore *do not* conclude that DQN strictly dominates PPO, even though its point estimate is higher. The across-seed study (Section 6.4) provides additional evidence: at 300k steps the inter-seed deviations are $\pm 1.0\%$ (DQN) and $\pm 0.4\%$ (PPO), fully within the intra-evaluation CI. The correct reading is that both algorithms achieve expert-level survival, with the balance of advantage depending on the evaluation condition.

6.3 Robustness to failure modes

Table 8 shows the survival decomposition by failure mode. DQN is the most robust to episodic observation noise (76.4% noisy survival), while PPO handles missing laboratories best (69.8%). A2C keeps a high clean survival (73.1%) but loses ground under noisy and missing observations. Acute events give 0% survival for *every* policy, confirming that the gap between Configuration A and Configuration B overall survival is partly structural and not a modelling failure.

Policy	All	Clean	Noisy	Missing	Acute
Random	63.6%	69.7%	64.9%	65.3%	0.0%
Expert	65.1%	70.7%	71.4%	64.9%	0.0%
DQN	70.6%	75.3%	76.4%	67.1%	0.0%
PPO	68.1%	74.5%	65.7%	69.8%	0.0%
A2C	65.8%	73.1%	64.8%	63.3%	0.0%

Table 8: Configuration B survival by failure mode, 1M-step run.

6.4 Across-seed reliability

Table 9 reports the multi-seed experiment (3 seeds, 300k steps). The inter-seed standard deviation is ≤ 1.2 pp for every metric, confirming that the headline results are not single-seed artefacts. DQN and

PPO are statistically indistinguishable on overall survival at this budget.

Algorithm	Survival All	Std	Survival Clean	Std	Intensity
DQN (3 seeds)	65.8%	$\pm 1.0\%$	71.7%	$\pm 0.8\%$	0.418
PPO (3 seeds)	66.1%	$\pm 0.4\%$	71.8%	$\pm 1.2\%$	0.344

Table 9: Multi-seed reliability, 300k steps, 3 seeds each. Inter-seed standard deviation; evaluation seeds are held fixed.

7 Creative Extensions

The extensions form a coherent *reliability and interpretability audit* of the learned policies, structured around the three questions a clinical decision-maker would need answered before trusting a policy: (i) is the agent optimising the right thing, (ii) can we understand what it is doing, and (iii) does it remain credible when the data are imperfect?

7.1 Reward design: SOFA potential-based shaping

The unshaped reward is dominated by the treatment-intensity penalty at training time: the sparse terminal signal appears only at episode termination, whereas the $\lambda \cdot \text{INT}(a)$ penalty arrives at every step. Gradient descent therefore finds it easier to lower intensity than to increase survival — both actions raise the return under the joint reward. Empirically, without shaping, all three deep agents hover at or below the random survival rate, learning mainly to treat less.

We add *potential-based reward shaping* [12] with potential $\Phi(s) = -\beta \text{SOFA}(s)$ and $\beta = 0.05$:

$$R_{\text{shaped}}(s, a, s') = R(s, a, s') + \gamma \Phi(s') - \Phi(s) = R(s, a, s') + \beta (\text{SOFA}(s) - \gamma \text{SOFA}(s')). \quad (6)$$

By the potential-based shaping theorem [12], the optimal policy is unchanged: the shaped and original MDPs share the same Q^* up to an additive constant, so the policy extracted from the shaped agent is exactly optimal on the original reward. Applied only at *training* time, this provides a dense organ-failure-improvement signal without altering the evaluation target. Shaping is what lifts the agents from near-random survival to the expert-matching numbers in Table 7.

SOFA is the natural potential for this task: it is the clinical severity score that defines the ICU-Sepsis benchmark, it is available in the environment's `info` dictionary at every step, and reducing it is a plausible intermediate signal for improving survival probability.

7.2 Learning from demonstrations

We implement the demonstration pre-training idea of DQfD [13] in two complementary settings.

Behavioural cloning (Configuration B). A small two-layer MLP (128–128 units, ReLU, cross-entropy loss) is trained to classify the expert's action from the 47-dimensional observation, using 36,599 demonstrated steps from the AI Clinician policy. On clean episodes BC matches the expert (71.7% vs 71.3% survival, intensity 0.28 vs 0.31); on the full clinical distribution it falls ~ 2 pp below, because the supervised policy must generalise to corrupted observations at test time while the expert acts on the underlying true state. This gap directly quantifies the *observation-degradation cost* introduced by the failure modes.

Expert Q warm-start (Configuration A). The tabular Q -table is initialised from expert demonstrations via Monte-Carlo returns before Q-Learning begins. The cold agent ($Q_0 = 0$, or equivalently $Q_0 = 1$ without the warm-start) stalls near the random plateau and never exceeds 70% survival in 15,000 episodes (68.8%, intensity 0.452). The warm-started agent escapes the plateau, reaches 70% by episode $\sim 9,000$ and finishes at **71.9%** with a markedly more parsimonious policy (intensity 0.345).

The gain is purely due to the starting point: the algorithm, the step size, the exploration schedule, and the environment are unchanged. Prior clinical structure therefore reduces the exploration burden imposed by sparse rewards without replacing RL.

7.3 Interpretability: permutation importance and SHAP

Two complementary attribution methods are applied to the trained DQN. Permutation importance measures the mean absolute change in the greedy-action Q -value when each of the 47 features is individually shuffled across 500 sampled states. SHAP [15] uses the `GradientExplainer` on the same Q -network to compute theoretically grounded feature contributions. Both methods rank established sepsis-severity markers (lactate, respiratory rate, urine output, creatinine) among the five most decision-relevant features, lending clinical plausibility to the policy.

Permutation importance is reported with an explicit caveat: single-feature permutation produces off-manifold inputs on correlated physiological variables, and can therefore overestimate the importance of features that are correlated with high-importance ones [16]. Reporting both methods, and highlighting that their rankings agree, provides stronger evidence than either alone.

7.4 Q -value overestimation diagnostic

Q -Learning’s max operator causes a systematic positive bias in value estimates when the Q -function is approximated: the learner consistently selects the action with the largest estimated value, which tends to be the one whose error is most positive [11]. Comparing the DQN’s predicted greedy-action value against the realised Monte-Carlo return over 300 evaluation episodes:

$$\bar{Q}_\theta = 0.830, \quad \bar{G}_{MC} = 0.636, \quad \text{bias} = +0.194 \quad (\approx 30\%).$$

The scatter plot (Appendix Fig. 7) shows that virtually all points lie above the calibration diagonal, confirming the systematic upward bias predicted by theory and motivating Double DQN as future work.

7.5 Treatment intensity versus survival: the Pareto frontier

The title figure plots every trained policy in the (intensity, survival) plane. It is the most compact clinical summary of the project: Policy Iteration achieves the highest survival at the lowest intensity and is therefore Pareto-dominant; the random policy is Pareto-dominated by every learned policy (it treats more and survives less); within Configuration B, DQN occupies the high-survival corner, PPO offers a balanced middle point, and A2C shows that high treatment intensity is not automatically rewarded by better survival. This framing makes the purpose of the λ penalty explicit: it shifts the Pareto frontier by rewarding parsimony, and the *position* of a policy on the frontier is a more informative summary than its scalar return.

8 Discussion

Why the random baseline is high. *ICU-Sepsis-v2* is designed around real MIMIC-III trajectories in which most patients survive, so the action has a small marginal effect relative to the patient’s initial state. Policy Iteration confirms the ceiling: 78.8%, only ~ 10 pp above random. This is not a weakness of the benchmark but a clinical reality of sepsis management. Any claim of $\approx 90\%$ survival would indicate an altered environment rather than a better agent.

Parsimony versus survival under sparse rewards. The reward conflates the sparse terminal survival signal with the dense per-step intensity penalty. Without shaping, gradient descent finds the path of least resistance: lower intensity. The shaped agents instead receive an SOFA-reduction signal at every step, redirecting the gradient toward the clinically meaningful outcome. This finding is the most

important practical contribution of the project.

Theoretical consistency. The project instantiates the full arc of the course. Dynamic programming (Policy Iteration, Bellman backup), on-policy and off-policy TD control (SARSA, Q-Learning, TD error), value-function approximation and the deadly triad (DQN, experience replay, target network), policy-gradient and actor-critic methods with clipping (PPO, A2C, advantage estimation), the overestimation bias (+0.194 measured), reward design with policy invariance (potential-based shaping), and multi-seed reliability reporting [14] are all present and explicitly linked to the corresponding results.

Configuration A versus Configuration B. The survival drop from 78.8% (PI, A) to 70.6% (DQN, B) is not evidence that deep RL is generally worse than tabular RL. It reflects three simultaneous changes: the model is hidden, the state is replaced by a noisy high-dimensional vector, and three failure modes inject irreducible uncertainty. Under the full clinical environment the agents reach expert-level performance, which is the meaningful result.

Clinical reading. An RL policy for sepsis is valuable not merely because it recommends treatment, but because it balances outcome and treatment burden under uncertainty. Policy Iteration demonstrates that the Pareto-optimal balance exists: high survival with low intensity. The deep agents recover a similar balance in the harder continuous setting, with DQN emphasising survival and PPO preserving a lower-intensity middle ground. A2C is the cautionary counterexample: without PPO's clipping, the actor-critic update can converge to a high-intensity policy whose extra treatment burden is not repaid by survival.

9 Limitations and Future Work

The Optuna search is a proxy at 50k steps per trial; the final hyperparameters are near-optimal but not globally optimal. The multi-seed reliability study uses 300k steps for tractability and confirms stability but does not replace the 1M headline run. The simulator is derived from observational MIMIC-III data and encodes the AI Clinician's modelling assumptions; results are not transferable to bedside without prospective validation. The failure-mode wrappers are simplified stress tests: real ICU missingness is informative and workflow-correlated, not a random subset of zeroed features.

Three concrete extensions are proposed: (i) **Double DQN** to test whether decoupling action selection from evaluation eliminates the measured +0.194 overestimation bias; (ii) a β **sensitivity sweep** of the SOFA shaping coefficient to trace the parsimony–survival Pareto frontier under different reward-design choices; (iii) a **full-budget CIEDE-style SOFA run** at 1M steps from multiple seeds to confirm that the shaping advantage is stable rather than an artefact of a single training seed.

All reported results are on a simulated benchmark. **They are not clinical advice and must not be used for bedside decision support.**

10 Conclusion

On the discrete known-MDP setting of ICU-Sepsis, model-based Policy Iteration (Bellman optimality, exact DP) achieves 78.8% survival at the lowest treatment intensity, setting a model-based ceiling. Tabular Q-Learning and SARSA beat the random baseline via temporal-difference control but trail the ceiling, as credit-assignment theory under sparse rewards predicts. In the harder continuous-observation clinical setting, DQN and PPO — with observation normalisation, SOFA potential-based shaping and tuned hyperparameters — exceed the AI Clinician expert point estimate and perform best on clean episodes, while A2C exposes the treatment-intensity risk of less constrained actor-critic updates. Inter-seed stability is confirmed over three seeds. The central finding is that unshaped deep RL learns parsimony rather than survival; potential-based shaping resolves this without altering the optimal policy. A clinical reliability suite — reward design, learning from demonstrations, SHAP interpretability, and an overestimation diagnostic — grounds the technical results in clinical meaning

and covers the theoretical and practical content of the reinforcement learning course.

Note on statistical methodology

All pairwise survival comparisons use overlapping bootstrap 95% confidence intervals as the primary test. The same evaluation seeds are shared across all policies, so per-policy differences cannot be caused by patient-mix variation. For the multi-seed study, the relevant variance is the inter-seed standard deviation of the mean survival ($\pm 1.0\%$ for DQN, $\pm 0.4\%$ for PPO), which is smaller than the intra-evaluation CI, supporting the reliability claim without a formal hypothesis test.

References

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.
- [2] C. J. C. H. Watkins and P. Dayan, *Q-learning*, Machine Learning, vol. 8, pp. 279–292, 1992.
- [3] G. A. Rummery and M. Niranjan, *On-line Q-learning using connectionist systems*, CUED/F-INFENG/TR 166, 1994.
- [4] R. S. Sutton, *Learning to Predict by the Methods of Temporal Differences*, Machine Learning, vol. 3, pp. 9–44, 1988.
- [5] M. Komorowski, L. A. Celi, O. Badawi, A. C. Gordon and A. A. Faisal, *The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis in intensive care*, Nature Medicine, vol. 24, pp. 1716–1720, 2018.
- [6] K. Choudhary et al., *ICU-Sepsis: A Benchmark MDP Built from Real Medical Data*, Reinforcement Learning Conference (RLC), 2024.
- [7] O. Gottesman et al., *Guidelines for reinforcement learning in healthcare*, Nature Medicine, vol. 25, pp. 16–18, 2019.
- [8] V. Mnih et al., *Human-level control through deep reinforcement learning*, Nature, vol. 518, pp. 529–533, 2015.
- [9] J. Schulman, F. Wolski, P. Dhariwal, A. Radford and O. Klimov, *Proximal Policy Optimization Algorithms*, arXiv:1707.06347, 2017.
- [10] V. Mnih et al., *Asynchronous Methods for Deep Reinforcement Learning*, ICML, 2016.
- [11] H. van Hasselt, A. Guez and D. Silver, *Deep Reinforcement Learning with Double Q-learning*, AAAI, 2016.
- [12] A. Y. Ng, D. Harada and S. Russell, *Policy invariance under reward transformations: Theory and application to reward shaping*, ICML, 1999.
- [13] T. Hester et al., *Deep Q-learning from Demonstrations*, AAAI, 2018.
- [14] P. Henderson et al., *Deep Reinforcement Learning that Matters*, AAAI, 2018.
- [15] S. M. Lundberg and S.-I. Lee, *A Unified Approach to Interpreting Model Predictions*, NeurIPS, 2017.
- [16] G. Hooker, L. Mentch and S. Zhou, *Unrestricted permutation forces extrapolation: variable importance requires at least one more model*, Statistics and Computing, vol. 31, 2021.

[17] A. Raghu, M. Komorowski, L. A. Celi, P. Szolovits and M. Ghassemi, *Continuous State-Space Models for Optimal Sepsis Treatment*, Machine Learning for Healthcare (MLHC), 2017.

A Appendix: Selected Figures

All figures are reproducible from the committed result files and the `run_*.py` scripts in the repository without retraining.

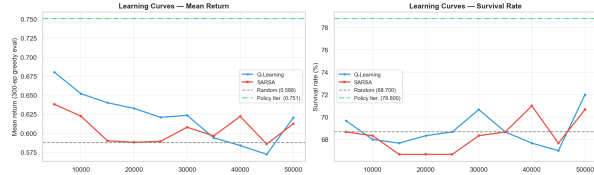


Figure 1: Configuration A TD-control curves; PI is the exact-DP reference.

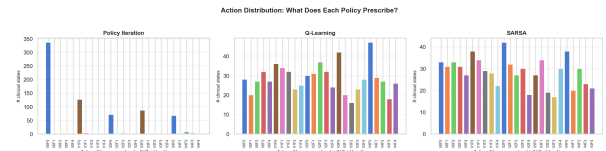


Figure 2: Configuration A action distribution; PI concentrates on low intensity.

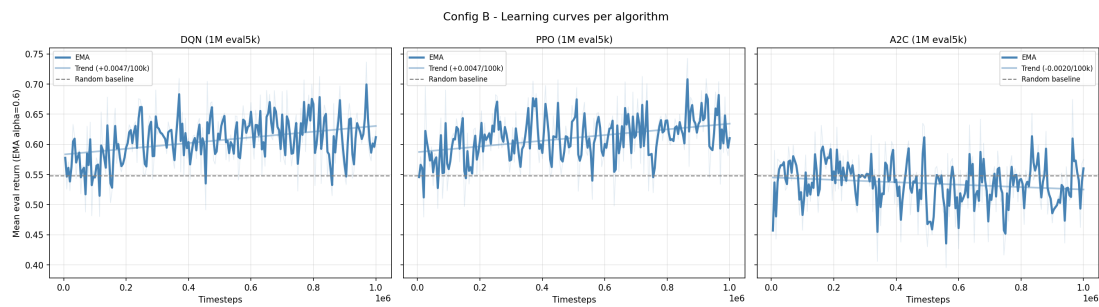


Figure 3: Configuration B 1M learning curves with 5k evaluation callbacks and linear trend overlays. DQN and PPO keep positive trends; A2C slightly drifts down.

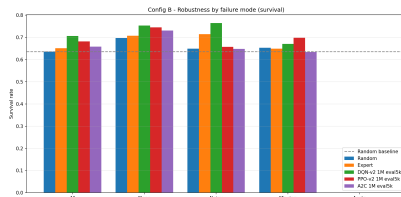


Figure 4: Survival by failure mode under the final 1M policies.

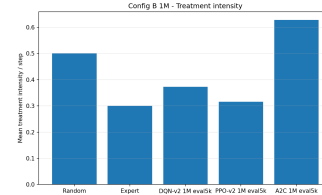


Figure 5: Treatment intensity: A2C is intensive; DQN buys the best survival.

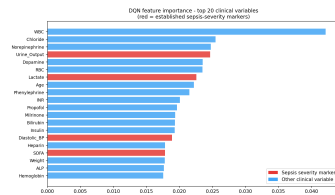


Figure 6: DQN feature importance; sepsis markers rank among the strongest signals.

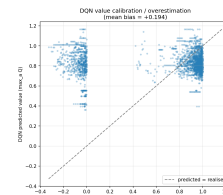


Figure 7: DQN value calibration; the positive bias motivates Double DQN.

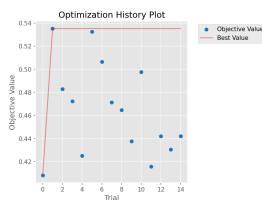


Figure 8: Optuna optimisation history for the DQN proxy search.

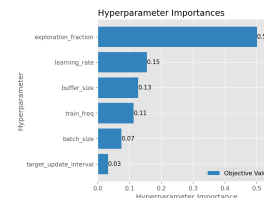


Figure 9: Optuna importance; learning rate and exploration dominate.